UNIVERSITY OF WOLLONGONG AUSTRALIA

ECTE471

Robotics and Flexible Automation Challenge

Group Project Assignment

Automated Assembly Cell — Competitive Engineering Tender

Group project: 4–6 students

Assessment: 100 marks

Professor Zengxi Pan

Teaching alignment: Lecture 01–10 and the ten-placement assembly task. Figure 1 gives approximate product views and Figure 2 fixes only the supply and assembly references. Teams choose the robot, base, grasp strategy, end-effector and TCP.

1. Client Brief — The Design Challenge

A manufacturer of small electric mobile vehicles is preparing a new automated assembly cell. The company has issued a competitive engineering tender and invited several robotics design companies to propose a solution.

Your group is one of these competing engineering design teams. Your objective is to design, analyse, simulate and convincingly present a technically feasible robotic system that can automatically assemble the supplied mobile robot.

Tender principle: The strongest proposal is not necessarily the most complicated. It is the proposal that best combines technical feasibility, engineering judgement, system integration, verification and professional communication.

2. Form Your Engineering Company

- Form a team of 4–6 students and choose a professional company/team name.
- Nominate a Project Manager / Team Coordinator.
- Allocate responsibilities for system design, kinematics, simulation, programming, advanced challenge and technical communication.
- All team members must understand the complete system, even when responsibilities are divided.
- A simple team logo or visual identity is optional.

3. Product and Assembly Requirements

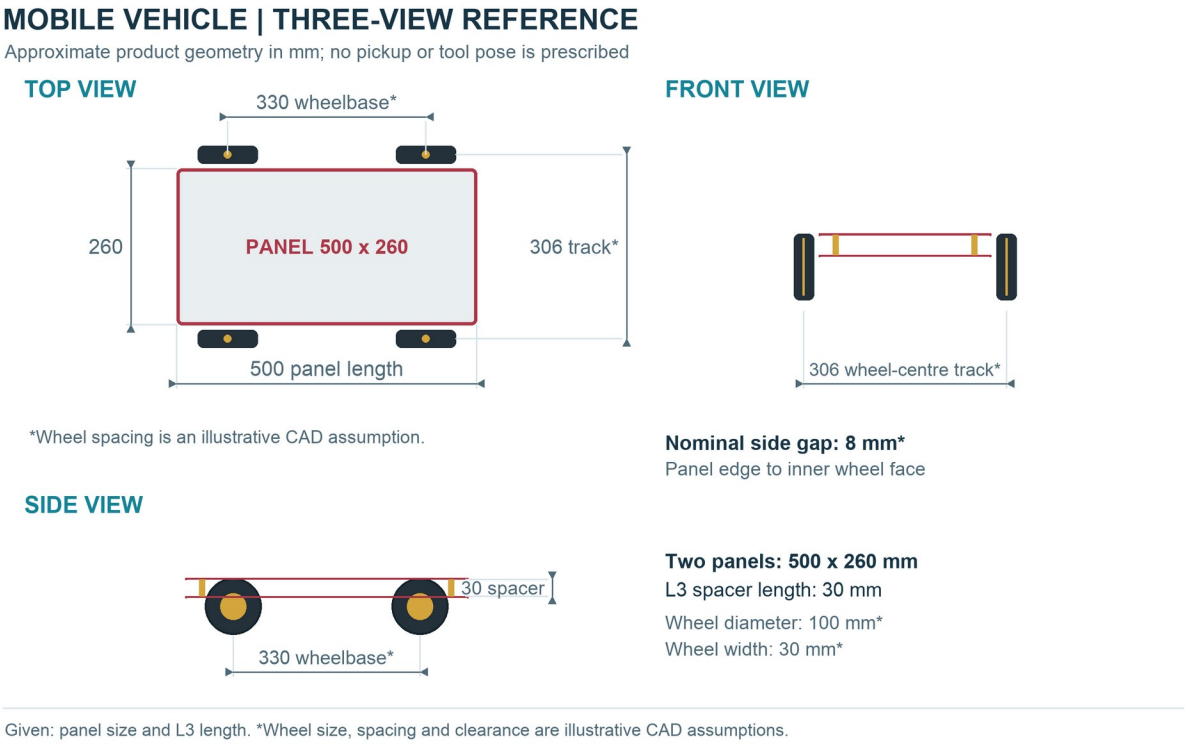


Figure 1. Approximate product top, front and side views. Panel size and spacer length come from the brief; starred wheel dimensions and clearances are illustrative CAD assumptions.

3.1 Supplied design data

- The robotic task contains ten placements: the lower panel, four wheels, four L3 spacers and the upper panel. The upper and lower panels are 260 mm wide × 500 mm long.
- The wheel motors are already installed on the lower panel before robotic assembly begins.
- The wheels are installed by pushing them onto the motor shafts.
- The L3 spacer is 30 mm long.
- End-effector geometry and contact points are team design choices. If a finger gripper is proposed, a 50 mm finger length may be used as an initial assumption and must be checked.
- Dimensions not provided may be reasonably assumed, but every important assumption must be stated and justified.

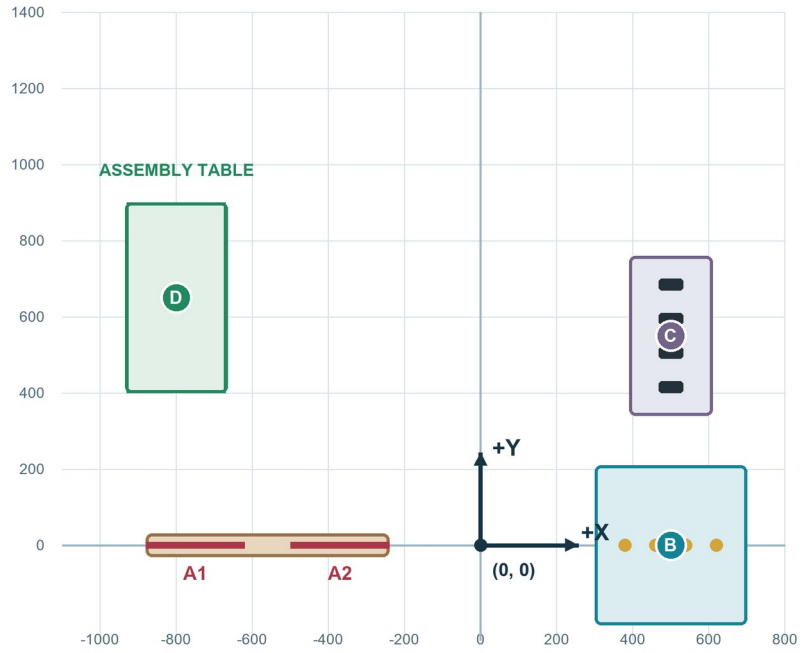
3.2 Robot and end-effector constraints

- Use one commercially available industrial robot with no more than six controlled degrees of freedom.
- Select a recognised manufacturer. No more than three groups may use robots from the same manufacturer.
- Use the Figure 2 references: two upright panel supply positions at X = −750 and −370 mm on the panel rack (Y = 0); spacer table centre (500, 0); wheel table centre (500, 550); and assembly product frame (−800, 650, 300) mm at 90° yaw. The team chooses and justifies the robot model and base position. No robot base, grasp point or TCP is prescribed by Figure 2. Use Tables 1 and 2 for the complete workcell levels and supply references.

- Select a suitable gripper/end-effector and actuation concept. Detailed finger motion need not be simulated.
- Choose each approach from the part and fixture geometry. Wheel insertion follows the motor-shaft axis; panel and spacer placement must preserve clearance and the required orientation.
- Do not include nuts, screws or fastening operations in the required ten-placement sequence. Choose and justify an end-effector strategy for the panels, wheels and spacers. Tool changes and a tool rack are optional design additions, not supplied or required facilities.

FIGURE 2 | STUDENT WORKCELL REFERENCES

Plan view in world X/Y; centres and footprints in mm



REFERENCE LOCATIONS

A1 / A2 Upright panels
X = -750 / -370; Y = 0
Rack centre (-560, 0); 640 x 60

B Spacer table
Centre (500, 0)
Footprint 400 x 420; four spacers

C Wheel table
Centre (500, 550)
Footprint 220 x 420; four wheels

D Assembly table
Product frame (-800, 650, 300)
Yaw 90 deg; table 270 x 500

Ten placements
Lower panel + 4 wheels
+ 4 spacers + upper panel

Only supply and assembly references are fixed. Robot, base, end-effector and pickup strategy are student choices.

Figure 2. Fixed supply and assembly references for the ten-placement task. The two upright panel positions, two parts tables and assembly table are shown; robot placement and grasp/TCP design are left to each team.

3.3 Workcell dimensions and supply references

All dimensions below are in mm in the Figure 2 world frame W, with +Z upward. The floor is Z = −450, so W's Z = 0 plane is 450 mm above the floor. Tables 1 and 2 supplement the Figure 2 plan view.

Table 1. Given workcell dimensions and surface levels

Station	Centre X, Y	Plan size X × Y	Surface Z	Above floor
A Panel rack	−560, 0	640 × 60	50	500
B Spacer table	500, 0	400 × 420	50	500
C Wheel table	500, 550	220 × 420	50	500
D Assembly support	−800, 650	270 × 500	300	750

A1 and A2 are upright in the XZ plane: the 500 mm panel dimension is vertical and the 260 mm dimension is horizontal. Their centres are at Z = 300; their lower and upper edges are Z = 50 and 550.

Table 2. Nominal supply arrangement and product reference

Supply / reference	X, Y	Z and orientation
A1 upright panel	−750, 0	Centre Z = 300; lower edge Z = 50
A2 upright panel	−370, 0	Centre Z = 300; lower edge Z = 50
B four spacers	X = 380, 460, 540, 620; Y = 0	Body lower ends at Z = 50; axes vertical
C four wheels	X = 500; Y = 415, 505, 595, 685	Wheels upright; axles parallel to Y; bottoms at Z = 50
D product frame	−800, 650	Z = 300; yaw = 90°

Table 2 gives the nominal arrangement shown in the supplied 3D layout. For the illustrative 100 mm wheel diameter in Figure 1, wheel centres are at Z = 100. If an unspecified component size is changed, state the resulting supply pose while retaining the given station references.

The D product frame is at the centre of the assembly support reference plane, at Z = 300 and yaw 90°. The 270 × 500 mm plan is a support envelope; detailed fixtures must allow clearance for the pre-installed motors and wheels. Robot/base selection, grasp points, TCPs and approach poses remain team design choices.

4. Tender Development Process

The project is developed through four engineering stages. EOI, PDR and FDR are assessed technical reviews, not formal class presentations. When a team is ready, it brings the specified summary slide(s) to an individual discussion with the lecturer. The review may be recorded as APPROVED, APPROVED WITH COMMENTS, or REVISE AND RESUBMIT.

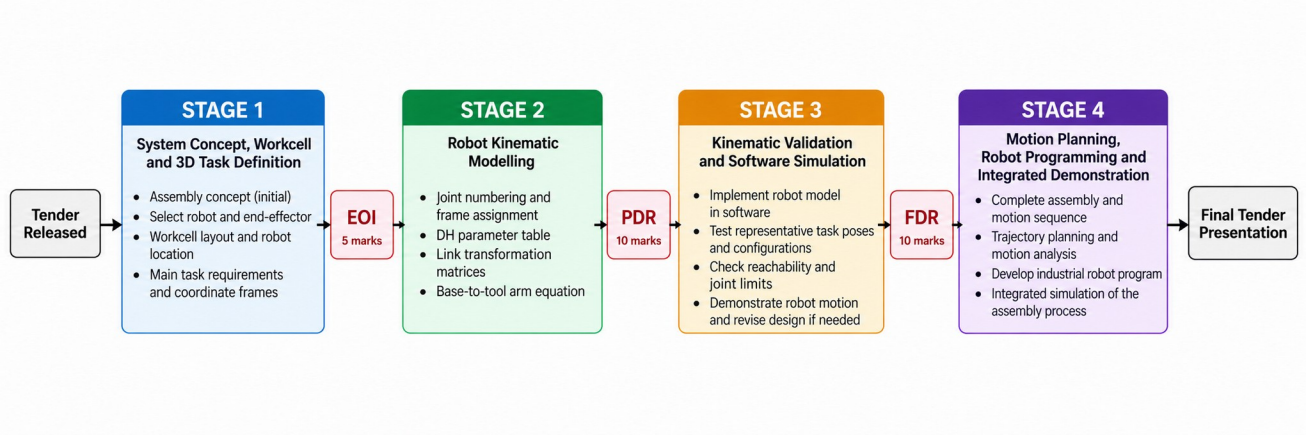


Figure 3. Four-stage tender and design-review process

4.1 Stage 1 — System Concept, Workcell and 3D Task Definition

1. Define the assembly sequence, components, task constraints and initial robotic concept.
2. Shortlist by workcell size, reach, required degrees of freedom and approach access. Then verify payload including tooling, load offset and joint limits against the selected manufacturer variant.
3. **Use the current C24 supply and assembly reference positions in Figure 2; propose and justify the robot model, base position and orientation.**
4. Define base, work-object, TCP and representative component/target frames. State required positions, orientations and approach directions.

EOI — 5 marks (end of Class 2 / after Lecture 2): Bring one summary slide showing the team/company, assembly/workcell concept, selected robot/model, DOF, payload, reach and concise suitability argument.

4.2 Stage 2 — Robot Kinematic Modelling

1. Number the joints and attach coordinate frames using a consistent DH convention.
2. Develop the complete DH parameter table and identify the variable parameter for every joint.
3. Derive representative neighbouring-link homogeneous transforms and the complete base-to-tool arm equation.
4. Check the mathematical frame directions against the physical robot.

PDR — 10 marks (end of Class 3 / after Lecture 3): Bring two to three slides showing the robot and joint numbering, frame assignment, DH table, one representative transform and the overall arm equation.

4.3 Stage 3 — Kinematic Validation and Motion Feasibility

1. Implement or import the selected robot in MATLAB or Python and compare the software model with the manual DH model.
2. Define representative pick, approach, place and retreat TCP poses, including orientation and configuration.
3. Run IK, compare solution branches, and check joint limits, workspace and physical plausibility.

4. Generate representative joint-space and Cartesian trajectories. Inspect position, velocity and acceleration and preserve the required task orientation.
5. **Revise the robot base, tool, grasp/target pose or motion choice when validation reveals a problem; keep the Figure 2 supply and assembly positions.**

If a chosen model cannot use ikine6s, follow Lecture 4: retain its real geometry and use a compatible analytical or numerical solver. Puma 560 is an allowed teaching fallback for IK exercises, but its results do not validate the selected project robot. Clearly separate fallback exercises from project evidence.

FDR — 10 marks (end of Class 5 / after Lecture 5): Bring two to three slides showing the software model, representative targets/configurations, reachability and joint-limit checks, basic trajectory evidence and at least one important modelling issue or correction.

4.4 Stage 4 — Programming, Advanced Challenge and Integrated Demonstration

1. Use the Lecture 5 eight-stage baseline: home, pick approach, pick and grip, pick retreat, assembly approach, assemble and release, assembly retreat, and return home. Add collision-avoiding via points and relevant grasp or release actions. A team may add tool changes if its own end-effector strategy requires them; fastening is outside the task.
2. Develop an industrial robot program. ABB RAPID is the course exemplar; an equivalent language may be used for the selected robot. Include targets, configurations, tool/work-object definitions, motion instructions, speed/zone choices and process placeholders.
3. Use the geometric Jacobian to interpret one demanding configuration or trajectory segment. State its frame and dimensions, and discuss joint-speed amplification, rank or conditioning. Distinguish angular velocity from Euler/RPY angle rates.
4. Complete one Advanced Engineering Challenge and integrate it into the same workcell and program.
5. Demonstrate the integrated cell in simulation. The result must provide engineering evidence, not only animation.

Use Lecture 10 cases to explain one existing integration decision, such as offline programming, calibration or inspection. Discuss what remains before commissioning. WAAM implementation, VLA training and an AI robot agent are not additional project requirements.

Final submission: Stage 4 concludes with the Final Engineering Report (35 marks) and Final Tender Presentation and Demonstration (40 marks).

4.5 Lecture-to-Project Map

Lecture	Project use
L1 Introduction	Robot and end-effector selection, specification interpretation, workcell concept and project launch
L2 3D Representation	Frames, pose, target orientation and transform chains
L3 DH Modelling	Robot model, forward kinematics and model checking
L4 Inverse Kinematics	Reachability, solution and configuration choice, joint-limit checks

Lecture	Project use
L5 Trajectory Planning	Joint and Cartesian trajectories, zones and via points, motion validation
L6 ABB RAPID Programming	Industrial program structure, targets, data, motion instructions, speed and zones
L7 Jacobian, Force and Compliance	Velocity and singularity analysis, and Advanced Challenge B
L8 Machine Vision and Calibration	Camera models, intrinsics, distortion, PnP and eye-in-hand calibration; target conversion for Advanced Challenge A
L9 Robot Path Planning	Optional support for collision-free route planning and Advanced Challenge C; detailed algorithms are not required of every team
L10 Lean Automation	Industrial context for offline programming, calibrated sensing, flexible tooling, process monitoring and validation; supports the final justification

5. Advanced Engineering Challenge — Choose One

Each team must select one challenge and develop it as a substantial extension of the same robotic cell. The challenge contributes 10 marks: 5 marks in the report and 5 marks for communicating the challenge in the presentation/demonstration. The work must define the extra client problem, use appropriate analysis or modelling, modify the design/program, provide evidence and discuss limitations.

5.1 Option A — Machine Vision and Robot Calibration

Client change: one component may vary by ±40 mm in X, ±40 mm in Y and ±15° in orientation.

- Define camera, robot-base, work-object, TCP and component frames.
- Distinguish camera/lens characterization (intrinsics and distortion), target-to-camera pose estimation and the fixed robot–camera transform. Explain TCP calibration separately; measure or define the workobject/fixture datum as needed.
- For a fixed camera, compose the calibrated camera-to-base transform with the measured object-to-camera pose. Include the object-to-TCP grasp offset to obtain the commanded TCP pose. For an eye-in-hand camera, include the current base-to-flange pose and the fixed camera mounting transform.
- Demonstrate updated pick poses for at least two component locations and run IK for each.
- Discuss feature localisation, lens distortion, calibration-pose error and the TCP/fixture uncertainty at the actual task height. Reprojection error in pixels alone does not prove millimetre accuracy at the robot target.

A sophisticated image-processing algorithm is not required. Simulated measurements are acceptable; the emphasis is calibration and perception-to-robot integration.

5.2 Option B — Force and Compliance in Robotic Assembly

Client change: wheels are occasionally damaged during insertion onto the motor shafts.

- Explain why pure position control may create excessive contact force and use $F = K\Delta x$ to support the reasoning.

- Select a force/torque sensor, passive compliance, compliant tooling, active force control or a justified combination.
- Define an approach → contact detection → reduced-speed insertion → force/displacement monitoring → success/fault → retreat sequence.
- State reasonable contact, success and excessive-force criteria and modify the program/control logic accordingly.
- Discuss practical benefits, limitations and implementation.

5.3 Option C — Collision-Free Robot Path Planning

Client change: a new fixture or obstacle blocks the original transfer path.

- Add realistic obstacle geometry and show why the original route is unacceptable.
- Develop a collision-free route using justified via points or an automated planner such as PRM/RRT.
- Check every robot link and the tool, not only the TCP; include safety margin, joint limits, orientation constraints and singularity risk.
- Compare at least two paths or robot configurations and justify the final choice.
- Demonstrate the validated path in simulation and then time-parameterise it for execution.

Lecture boundaries: the latest L8 image-processing overview and recent robotic-vision developments are optional, and Part 7 is not examined. Simulated pose measurements remain acceptable for Option A. L9 is an optional learning module that supports Option C; justified, fully checked via points remain an acceptable solution.

6. Computational Tools and Responsible AI Use

6.1 MATLAB or Python

- Either MATLAB or Python may be used; the engineering standard and marking are identical.
- Teams must select appropriate libraries, implement calculations, test code and resolve language-specific issues.
- Submitted code must run and reproduce the reported results. Students may be asked to explain or modify code during review.
- RobotStudio, RoboDK, Webots or manufacturer software may supplement, but not replace, the required engineering analysis.

6.2 Generative AI

- AI may assist brainstorming, concept explanation, coding/debugging and feedback.
- All AI-assisted code, equations, specifications, references and outputs must be checked and understood by the team.
- The final report must be written by the team in its own words and reflect its own reasoning. Do not submit substantial AI-generated or AI-rewritten passages, consistent with Lecture 1.
- Include an AI Use Declaration identifying the tool, purpose, where it assisted and how outputs were verified.
- Keep drafts, version history and important development records; these may be requested to confirm authorship.

AI Use Declaration field	Required information
Tool(s)	Name the AI system(s) used
Purpose	Brainstorming, explanation, coding/debugging, feedback or another permitted purpose
Where AI assisted	Identify project stages, code modules or tasks
Verification	Explain how calculations, code, references and outputs were checked
Team declaration	Confirm the report text is team-written and all members can explain the work

7. Deliverables

7.1 Final Engineering Report

- Executive summary and design assumptions
- Assembly sequence, robot/end-effector selection and workcell design
- Coordinate frames, TCP/target pose definitions and transformation chains
- DH model, forward kinematics and software-model validation
- Inverse kinematics, reachability, configuration and joint-limit analysis
- Trajectory generation, joint velocity and Jacobian/singularity analysis
- Industrial robot program and system integration
- Selected Advanced Engineering Challenge
- Integrated simulation, evidence status, limitations, improvements and conclusions, including an industrial integration observation from Lecture 10
- Group contribution statement and AI Use Declaration

7.2 Simulation and Code

Upload all MATLAB/Python scripts and relevant simulation/program files before presenting. Files must be organised, documented and runnable so major results can be reproduced.

7.3 Final Tender Presentation and Demonstration

Approximately 15 minutes plus 3–5 minutes of technical questions. Three to six members may present, but all members must be ready to answer questions.

- Client problem and proposed robotic system
- Key engineering decisions and justification
- Technical validation and selected advanced challenge
- Integrated simulation/demonstration
- Limitations and improvements
- Final recommendation: why should the client select your design?

8. Assessment Criteria

Project-development review	Marks	Strong evidence
EOI	5	Credible concept, appropriate robot proposal, key specifications and concise justification
PDR	10	Correct frame assignment, DH table, representative transform and complete arm model
FDR	10	Working software model, feasible targets/configurations, joint-limit checks and basic trajectory evidence
Total	25	

Final Engineering Report	Marks	Strong evidence
System concept, robot/end-effector and workcell	5	Coherent, practical and supported by manufacturer/task data
Pose representation, kinematics and validation	8	Correct frames/transforms/DH/FK/IK with physical interpretation
Trajectory, Jacobian, programming and integration	10	Appropriate motion analysis, singularity interpretation, program and integrated evidence
Engineering justification and report quality	7	Complete detail, explicit assumptions, clear reasoning and professional team-written communication
Selected Advanced Engineering Challenge	5	Meaningful analysis, integration, evidence and limitations
Total	35	

Assessment totals 100 marks: Development Reviews 25 (EOI 5, PDR 10, FDR 10), Final Engineering Report 35, and Final Tender Presentation and Demonstration 40. Quizzes and the optional L9 laboratory are separate exercises and add no marks to this group-project total. The report assesses technical work; the presentation assesses how clearly the team explains, demonstrates and defends it.

Final Tender Presentation and Demonstration	Marks	Strong evidence
Structure and audience-focused narrative	10	Clear purpose, logical sequence, useful transitions and an audience-appropriate conclusion.
Delivery, visuals and time management	10	Clear speech, readable visuals, effective pacing and professional delivery within the allocated time.
Explanation of the integrated demonstration	10	Explain the sequence, evidence and limitations clearly; 5 of these 10 marks concern communication of the selected advanced challenge.
Q&A and coordinated team response	10	Listen to questions, give clear and accurate answers, explain decisions and involve the team effectively.
Total	40	

9. Professional Expectations

- Use clear figures, labels, units, coordinate-frame notation and matrix dimensions.
- Make assumptions explicit and explain why important decisions were made.
- Use manufacturer data and reproducible simulation evidence where appropriate.
- Check orientation, configuration, joint limits, collision and singularity—not only TCP position.
- Design iteration is expected; discovering and correcting a problem is evidence of engineering practice.

Professor Zengxi Pan

3 October 2026